AURIGENE

PHARMACEUTICAL SERVICES

Astellas collaboration - Competitive Rate Card

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1. Background

Aurigene Pharmaceutical Services Limited (APSL) has progressed in Astellas's strategic research supplier pool selection process for routine chemistry services and routine biologics services. Additionally, Astellas is seeking competitive rate cards. This document is a summary in response to the request.

2. Business Proposal

APSL proposes an FTE arrangement with flexibility in deploying FTEs according to the project's needs. Additionally, for certain services that can operate independently, a fee-for-service (FFS) pricing model has been provided for added flexibility. This proposal will offer Astellas options to engage in either a fully FTE mode or a hybrid (FTE+FFS) mode for different services at different stages of operations. A few salient features of the proposed collaboration are listed below:

a) Project group will have a PhD in relevant area of expertise with industrial R&D experience, excellent in English speaking and writing, good project management skills, strong technical trouble shooting capability.

b) Additionally, APSL will deploy a project manager, fluent in English, to co-ordinate operational aspects of the project.

c) Deployment of FTEs at any phase will be finalized after discussion and approval by Astellas.

d) When projects are executed in FFS mode, APSL will deploy scientists with relevant skill set to execute on the deliverables as per agreed timelines. For each FFS project a detailed proposal, taking into account project specific nuances will be submitted with appropriate timelines and pricing details.

The scope of work that will be undertaken in the proposed FTE model will include the following activities:

ACTIVITIES	**DETAILS**
Chemistry activities (Small molecule)	Synthetic route scouting of compounds designed by Astellas, on routes suggested by Astellas
	Preparation of compounds in mg to multi gram scale, purity >95%
	Racemic final compounds will be resolved by chiral HPLC if specified by Astellas
	Analytical characterization of all synthesized compounds
	Shipment by certified express carriers in accordance with DAP Incoterm. APSL will be responsible for all the documentations for shipment.
	Experimental procedures fully documented in notebooks
	Experimental procedures reported in ACS publication format
	Weekly status reports

Analytical chemistry	All final compounds will be characterized by HPLC, LCMS and NMR; data will be provided to Astellas weekly for each final compound ready to be shipped to a specified location

Biologics	Routine Biologics Services include, but are not limited to:
	• DNA synthesis services (for multiple types of DNA such as single/double strand DNA and liner/circular DNA)
	  • DNA synthesis
	  • DNA cloning to suitable plasmid vector
	  • DNA encapsulation to suitable virus vector (AAV)
	  • DNA purification and analysis for QC
	• RNA synthesis services (for multiple types of RNA such as mRNA)
	  • Template DNA synthesis
	  • RNA synthesis
	  • RNA purification and analysis for QC
	• Protein synthesis services (for multiple types of protein such as natural protein, antibody, peptide, etc.)
	  • Protein production using cell culture or cell-free systems
	  • Protein purification form cell culture medium
	  • Protein analysis for QC

Biology services	Toxicology, Pharmacology, and ADME/Biodistribution studies
	DMPK FFS rate card is attached in Annexure 1
	Based on study design, Pharmacology and toxicology rates will be shared with Astellas for further discussion.
Ad-Hoc Report Writing	Report writing will be billed additionally based on the type of report.

2.1 Quality Parameters/ SLAs

Infrastructure Requirements: APSL will provide dedicated state of the art labs with access to consumables and dedicated instrumentation.

- APSL' s current infrastructure comprises over 50 chemistry labs of which the appropriate labs can be readily assigned to the Astellas project. APSL will allocate a DEDICATED LAB for every 10 FTEs.
- The modular set-up of chemistry labs includes the following instrumentation per lab:
  - 5-10 fume-hoods with supply of chilled water, nitrogen, air and vacuum
  - ~ 10 or 22 stirrers
  - 2-4 rotary evaporators
  - 1 oven
  - 3 high vacuum pumps
  - Grids for fitting flash column in each fume hood
  - Dedicated MPLC purification machine
- APSL's biologics services are provided through state-of-the-art laboratories spanning 10,000 square feet, featuring dedicated suites for mammalian, viral, and bacterial work streams, each equipped with individual air handling and safety systems. Our labs and processes are designed for low endotoxin levels in the deliverable.

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APSL will also be responsible for providing sufficient other personnel as necessary to perform all administrative and other non-scientific activities to support the services.

Also, as an added advantage, APSL has a chemical inventory system that has an inventory of about 50,000 chemicals. This inventory system is managed with barcode for all 50,000 chemicals and about 10,000 chemicals are stored at cold storage. This inventory system provides an efficient way to manage chemicals (effective utilization of chemicals) thereby helping scientists to jump start the chemistry ASAP.

Similarly, for Biology and Biologics services, APSL has a repository of standard ATCC cell lines, commercially sourced vectors and host cells which can be used for R&D purpose. Cell lines and vectors, not available in our inventory, can be procured with sufficient notice.

Productivity metrics will be closely tracked and best commercial efforts to achieve the desired indicative output metrics upon discussion with Astellas.

Sample KPI Template (as may be modified by Astellas):

		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
Total FTE's														
Shipped Compounds	**>1 g**													
	100 mg – 1g													
	10 mg - 100 mg													
	1-10 mg													
Total No. of Compounds Shipped														
Total Amount of Compounds Shipped (gm)														
Avg. amount of Compounds Shipped (gm)														
Compounds shipped/FTE														
Avg. Synthesis Time in Days														
Total No. of Steps completed														
Avg. No. of Steps/FTE														
Total No. of Reactions Performed														

Avg. No. of Reactions/FTE													
Reactions/Step													
Accident													
FTE Retention													

3. Project Pricing Summary (FTE):

FTE Pricing		
Function	**FTE Tier (#)**	**Annual FTE rate (USD)**
Chemistry#	1 - 25 26 - 50 51 - 100 >100	68,000 67,000 66,000 64,000
CADD	1 - 5	70,000
Biologics*	1 - 5 6 - 10 11 - 15 >15	85,000 82,000 79,000 77,000
Biology	1 - 5 6 - 10 11 - 15 >15	80,000 78,000 76,000 74,000
Other costs		
Pass-through costs	For chemistry FTEs: All consumables except the ones listed in Annexure-1. For Biology and Biologics FTEs: All consumables All low temperature shipments High resolution mass spec and SPR usage will be charged on hourly basis. Details on the hourly rate is mentioned in Annexure-4 Consumables include columns, resins, animal sourcing, kits, reagents, fine chemicals etc.	
Chiral separation costs	Included in FTE pricing up to final product output of <25 mg scale. On need basis, a separate FFS quote will be shared with Astellas for chiral compounds with >50 mg of output.	
Scale up	Cost of raw materials, reagents, solvents and consumables for synthesis and purification of >1g output of final compounds is not included and will be billed at actuals	
Shipping costs-Normal shipment	One room temperature shipment per week to Astellas is included in the FTE Pricing. Additional shipments or temperature-controlled shipments will be billed at actuals	
Licensing fee	All licensing fees are pass-through. Alternatively, Astellas can opt to extend licenses to APSL for specific usage in Astellas' projects	

Doesn't include oligonucleotide synthesis services. FFS rate card is appended below in Annexure 3.

*Routine biologic services are also provided on rate card basis and the same is attached as Annexure 2. Biologics FTEs will be fungible and team members with the requisite skill sets will be allocated to the project as needed, ensuring efficiency and seamless delivery.

Direct Award Discounts:

Additional discount will be offered on the net purchase across all services as per the table below:

Total order in a calendar quarter (USD)	Between 250,000 and 500,000	Between 500,000 and 1,000,000	Greater than 10,000,000
Discount	2%	3.5%	5%

Illustration of discount calculation**:** In a particular quarter, if the total order is USD 400,000, then for the first USD 250,000, no discount will be considered. For the value between USD 250,000 and USD 400,000, 2% discount will be considered. Total discount will be 2% of USD 150,000 which is USD 3,000.

4. Terms and conditions

FTE Price Validity and Invoicing

a) The pricing is valid for period of one year from start date. APSL will revisit the FTE rates annually which shall be based on consumer price index of India and INR-USD forex rate correlation.
b) FTE billing will be done, for the actual number of FTEs agreed between Astellas and APSL during the said period, on monthly basis.
c) The invoice for all pass-through expenses will be raised at the end of each month and same shall be paid by Astellas on monthly basis.

Payment Terms:

d) All pass-through costs will be charged at actuals, evidence to support the cost shall be provided, if requested.
e) APSL shall invoice Astellas for the agreed upon FTEs during the said period monthly, on the 5th of the month for the last month. For example, January invoice shall be sent in PDF by February 5th, and a hard copy sent by courier without any delay.
f) The invoice for consumables and shipping costs at actuals will be raised monthly, on the 5th of the month for the last month. APSL shall seek pre-approval from Astellas for such reimbursable consumables.
g) All Payments will be due within sixty (60) days/ as per MSA term from the date of invoice.
h) Payments for reimbursable will be invoiced monthly

5. Communication, Reports and Update Schedule

The formats of all reports will be mutually decided by the Project leaders according to the internal needs of Astellas (Final report template, SEND data sets, etc).
Weekly short reporting will be established to ensure a steady flow of information to Astellas.

The full report and the KPIs will be delivered one week after end of month.

- Regular web/ Telecon (TCs) with project leaders and responsible scientists from Astellas will be scheduled routinely as mutually agreed between Astellas and APSL scientific teams, co-ordinated by a Project Manager.
- Reporting and TCs will be scheduled in a way that the reports are available 3 days in advance of the TC.
- At any time, questions and potential issues will be addressed by email and by direct phone calls.
- Astellas will provide a share point for data storage and transfer. APSL can also support in generation of dedicated share point for this program on need basis, if suggested by Astellas.
- Astellas will provide 2-3 weeks advance information to APSL about forthcoming projects to help APSL complete raw material procurement and other project scheduling related activities.

6. Documentation

Experimental information will be entered into Astellas notebooks to be provided by Astellas and in full accordance with the Astellas notebook policy and SOP or as mutually agreed between Astellas and APSL.
Astellas will provide APSL the SOP 'Instructions for use of laboratory notebooks' and APSL guarantees that all involved FTEs observe the guidelines of the SOP.

7. Intellectual Property

a) APSL will ensure that Intellectual Property generated during the term of this project shall be transferred to Astellas. Astellas will exclusively own all data generated for the project(s).
b) APSL recognizes the value of Intellectual Property for the collaborators and takes necessary measures to secure the same in addition to measures taken in information flow and data management.

8. Other

Astellas may terminate the project with a notice period of thirty (30) days in case Astellas considers that the project within which Astellas entrusted APSL with the performance of the services is not of interest any more for any reasons or that following preliminary results obtained during the performance of the services holds no longer any interest to have the services performed by APSL.

In case of termination for any cause whatsoever, APSL shall (i) return to Astellas all data, documents and remaining quantities of materials, samples, intermediates and product supplied, (ii) provide Astellas with a report concerning the services as well as the results already obtained at the effective date of termination. Astellas shall pay to APSL the sums due in consideration of the services actually performed and for reimbursable expenses reasonably incurred up to the effective date of termination, including unavoidable costs for product-specific reagents already ordered or otherwise committed at time of termination.

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All materials including any compounds, protocols, data, reagents etc. that are transferred by Astellas to APSL during the projects will be returned to Astellas once the project is completed. All deliveries from APSL will declare the correct value of goods in the customs declaration. Any claims from the customs authorities would go to APSL's own expense and fall under APSL's responsibility/liability. APSL will retain data, records, biospecimens and samples for a period of at least 5 years following completion of the work.

9. Annexure 1

9.1 Standard Laboratory Chemicals for chemistry FTEs

Chemical	Specification	CAS No
Alumina, Neutral	Aluminium oxide 90, 70-230 mesh ASTM	1344-28-1
Ammonia solution, 25%	28.0-30.0% as NH3	7664-41-7
Ammonium chloride	≥99.5%	12125-02-9
Ammonium hydroxide		1336-21-6
Anti-bumping granules		
Calcium chloride	technical grade, −4-+30 mesh	10043-52-4
Celite 545		61790-53-2
Activated Charcoal	decolorizing agent	7440-44-0
Citric acid	≥99%	77-92-9 99
Cuprous chloride	97%	7758-89-6
Drierite, 8 mesh, no indicator	without indicator, 8 mesh	7778-18-9
Hydrobromic acid		10035-10-6
Hydrochloric acid (sg 1.18)	37%	7647-01-0
lithium hydroxide	≥98%	1310-66-3
Magnesium sulfate	anhydrous, reagent grade, ≥97%	7487-88-9
Molecular sieves		70955-01-0
Ninhydrin spray		485-47-2
Nitric acid	70%	7697-37-2
Phosphomolybdic acid	95%	51429-74-4
Phosphoric acid		7664-38-2
Phosphorus pentoxide	≥98.0%, powder	1314-56-3
Potassium acetate		127-08-2
Potassium bicarbonate		298-14-6
Potassium carbonate	99%	584-08-7 99
Potassium fluoride	≥99.0%	7789-23-3
Potassium hydroxide	≥90%, flakes	1310-58-3
potassium phosphate tribasic		7778-53-2
potassium phosphate dibasic		7758-11-04
potassium phosphate monobasic		7778-77-0
Sand	−50-70 mesh particle size	14808-60-7
Sieves, 3 angstrom	pellets, 1.6 mm	308080-99-1
Sieves, 4 angstrom powder	powder, activated, 5 µm	70955-01-0
Sieves, 4 angstrom	pellets, 1.6 mm	70955-01-0
Silica gel with moisture indicator (blue gel), dessicant ~1-3mm		112926-00-8
Silicone oil, high temp. (230 C)	Silicone oil, high temp. (230 C)	63148-62-9
Sodium acetate		127-09-3
Sodium bicarbonate	≥99.5%, crystalline	144-55-8
Sodium carbonate	≥99.0%	497-19-8
Sodium chloride (AnalaR)	≥98%, 80 mesh particle size	7647-14-5

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Chemical	Specification	CAS No
Sodium hydride, 60% in oil	60 % dispersion in mineral oil	7646-69-7
Sodium hydroxide	97%, beads	1310-73-2
Sodium sulfate	≥99.0%, anhydrous, granular	7757-82-6
Sodium sulfite		7757-83-7
Sodium thiosulphate.5H2O	Pentahydrate, 99.5%	10102-17-7
Sodium iodide	≥99%	7681-82-5
Solka floc		
Sulfuric acid	95-98%	7664-93-9
Acetic acid	glacial, 99.8%	64-19-7
Acetone	≥99%	67-64-1
Acetonitrile (LC/MS)	LC-MS grade, ≥99.9%	75-05-8
Acetonitrile	99%	75-05-8
Chloroform	≥99%	67-66-3
Cyclohexane		110-82-7
1,2-dichloroethane	≥99.0%	107-06-2
Dichloromethane	≥99.5%,	75-09-2
Diethyl ether, AR	≥99.0%	60-29-7
Diisopropyl ether		108-20-3
N',N-dimethylacetamide		127-19-5
N,N-Dimethylformamide	≥99.8%	68-12-2
Dimethylsulfoxide	≥99.9%	67-68-5
1,4-dioxane	≥99.0%	123-91-1
Ethanol	reagent grade, denatured	64-17-5
Ethyl acetate	≥99.5%,	141-78-6
n-Heptane	99%	142-82-5
Hexanes		110-54-3
Methylcyclopentane		96-37-7
3-Methylpentane		96-14-0
2-Methylpentane		107-83-5
Isopropyl acetate		108-21-4
Methanol	≥99.5%,	67-56-1
Methanol (for LCMS)	LC-MS grade, 99.9%	67-56-1
Methyl tert-butyl ether		1634-04-4
2-propanol	≥99%	67-63-5
Tetrahydrofuran	≥99.0%, Inhibitor-free	109-99-9
Toluene	≥99.5%	108-88-3
Anisole	99%	100-66-3
Triethylamine	≥99%	121-44-8
Pyridine	≥99%	110-86-1

9.2 DMPK Assays Price List

In vitro Studies			Price per Comp ($)		
Protocol category	**Protocol**	**Assay Description**	**1-9 Comps**	**10-24 Comps**	**>25 comps**
Stability in liver microsomes (MetCl)	MetCl-Human	n=1; 4 sampling timepoints; t ½; Cl int	140.00	107.00	97.00
	MetCl-Rat		140.00	107.00	97.00
	MetCl-Mouse		140.00	107.00	97.00
	MetCl-Dog		140.00	107.00	97.00
	MetCl-Monkey		174.00	133.20	120.00
	MetCl-Minipig		140.00	107.00	97.00
Hepatocyte clearance	Hepatocytes-Human	n=1; 6 sampling timepoints; t ½; Cl int	394.00	302.40	273.00
	Hepatocytes-Rat		300.00	230.40	208.00
	Hepatocytes-Mouse		300.00	230.40	208.00
	Hepatocytes-Dog		337.00	259.20	234.00
Plasma Stability	PlasmaStab-Human	n=1; 6 sampling timepoints; t ½	203.00	156.00	141.00
	PlasmaStab-Rat		211.00	162.00	146.00
	PlasmaStab-Mouse		211.00	162.00	146.00
	PlasmaStab-Dog		211.00	162.00	146.00
	PlasmaStab-Monkey		227.00	174.00	157.00
	PlasmaStab-Minipig		227.00	174.00	157.00
S9 Stab	S9Stab-Human	n=1; with & without Co-factors; 6 sampling timepoints; t ½	195.00	150.00	135.00
	S9Stab-Rat		195.00	150.00	135.00
	S9Stab-Mouse		195.00	150.00	135.00
	S9Stab-Dog		195.00	150.00	135.00
	S9Stab-Monkey		219.00	168.00	152.00
	S9Stab-Minipig		219.00	168.00	152.00
Glucuronide Conjugation using liver microsomes (MiGluc)	MicGlu-Human	n=1; Incubation in liver microsomes with cofactors NADPH & UDPGA; 4 sampling timepoints; t ½;	186.00	143.00	129.00
	MicGlu-Rat		186.00	143.00	129.00
	MicGlu-Mouse		186.00	143.00	129.00
	MicGlu-Dog		186.00	143.00	129.00
	MicGlu-Monkey		229.00	176.00	159.00
	MicGlu-Minipig		186.00	143.00	129.00
PPB	PPB-Human	n=3; Equilibrium dialysis; 4 h incubation; % free; fu; % recovery	224.00	171.60	155.00
	PPB-Rat		224.00	171.60	155.00
	PPB-Mouse		266.00	204.60	185.00
	PPB-Dog		266.00	204.60	185.00
	PPB-Monkey		327.00	250.80	226.00
	PPB-Minipig		309.00	237.60	214.00

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In vitro Studies			Price per Comp ($)		
Protocol category	**Protocol**	**Assay Description**	**1-9 Comps**	**10-24 Comps**	**>25 comps**
Microsomal Binding	Human	n=3; Equilibrium dialysis; 4 h incubation; % free; fu;	395.00	303.60	274.00
Tissue Binding	Brain Binding (Rat)		258.00	198.00	179.00
	Testes Binding (Rat)		294.00	225.50	203.00
TDIkobs	CYP3A4 TDI kobs	n=1, kobs, pre-incubation: 0, 5, 15, 30 min	156.00	120.00	108.00
Reversible CYP Inhibition	CYP3A4-Midaz	n=1; IC_{50}; 6 conc	164.00	126.00	114.00
	CYP3A4-Testosterone		164.00	126.00	114.00
	CYP2D6		164.00	126.00	114.00
	CYP2C9		164.00	126.00	114.00
	CYP2C19		164.00	126.00	114.00
	CYP2C8		164.00	126.00	114.00
	CYP1A2		164.00	126.00	114.00
	CYP2B6		164.00	126.00	114.00
Solubility	Thermodynamic Solubility	n=1; 1 Test Conc, 24 h incubation	115.00	100.00	80.00
	HTS Solubility	n=3; 18 h incubation	97.00	74.00	60.00
	Kinetic Solubility	n=3; 18 h incubation	102.00	78.00	71.00
Log D	Log D	n=3	141.00	108.00	98.00
Chemical Stability	Stability in Buffer	n=1, buffer pH 7.4, 24h incubation at room temp & 37 °C	173.00	150.00	135.00
MDCK	MDCK (native cells)	n=3; unidirectional transport; Pe(app)	147.00	113.00	102.00
	MDCK-BCRP (multiples of 5 comps per study)	n=2; 60 min bi-directional transport	962.00	740.00	592.00
	MDCK-MDR1 (multiples of 5 comps per study)	n=2; 60 min bi-directional transport	962.00	740.00	592.00
PAMPA	PAMPA	n=3; unidirectional transport; Pe(app)	124.00	95.00	86.00
Caco-2	Bi-directional transport	n=2; 60 min bi-directional transport	605.00	465.00	419.00
	Transporter Assessment (P-gP)	n=2; 60 min bi-directional substrate & inhibitor assessment	994.00	764.00	688.00
	Transporter Assessment (BCRP)	n=2; 60 min bi-directional substrate & inhibitor assessment	994.00	764.00	688.00

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In Vivo Studies			Price per Study ($)	
Protocol category	**Protocol**	**Study Description**	**1-9 Studies**	**>10 Studies**
Rodent PK	PK Discrete (1 Comp, one-route)	Rat or mouse; n=3; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	2,500.00	1,620.00
	PK Discrete (1 Comp, two-routes)	Rat or mouse; bio availability; n=3; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	3,600.00	2,400.00
	PK cassette (2-6 Comps, one-route)	Rat or mouse; 2-6 compounds in cassette; n=3; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	8,400.00	4,800.00
	PK cassette (2-6 Comps, two-routes)	Rat or mouse; 2-6 compounds in cassette;n=3: bio availability; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	11,550.00	6,600.00
	Ascending Dose PK	Rat or mouse; 2-4 doses; n=3; 8-12 timepoints; 1 Analyte; PK parameters	based on study design	
Rodent Brain Penetration	PK brain penetration discrete (1 Comp)	Rat or mouse; n=3; 1 route; 2 timepoints; brain & blood/plasma; 1 Analyte	2,730.00	1,560.00
	PK brain penetration discrete (1 Comp)	Rat or mouse; n=3; 1 route; 8 timepoints; brain & blood/plasma; 1 Analyte	5,688.00	3,250.00
	PK brain penetration discrete (1 Comp)	Rat or mouse; n=3; 1 route; 8 timepoints; brain, blood/plasma & CSF; 1 Analyte	7,700.00	4,400.00
	PK brain penetration cassette	Rat or mouse; 2-6 compounds in cassette; n=3; 2 timepoints; brain & blood/plasma; 1 Analyte	7,350.00	4,200.00

In Vivo Studies			Price per Study ($)
Protocol category	**Protocol**	**Study Description**	
Dog PK	PK Discrete (1 Comp, one-route)	Beagle dog; n=3; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	5,200.00
	PK Discrete (1 Comp, two-routes)	Beagle dog; bio availability in crossover design with one week washout period between doses; n=3; 8-9 timepoints up to 24 h; 1 Analyte; PK parameters	9,160.00

Note-

• The prices quoted for above are assumed to be performed in a screening mode as per standard APSL study design. Pricing may vary for custom study designs, based on client requirement.

• Data will be shared in a standard MS excel report.

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10. Annexure 2

Rate Card for regular biologics services (inclusive of consumables):

DNA synthesis and cloning:

S. No	DNA synthesis# and cloning	Price in USD for each construct (batch of 1 to 10)	Price in USD for each construct (batch of 10 to 50)	Price in USD for each construct (batch of 50 to 100)
1		290	255	220

#gene size assumed as 600 base pair

Protein expression screening:

S. No.	Expression system	Price in USD for small scale expression trials per protein
1	Microbial	1,500
2	Mammalian	3,000
3	Insect cells	4,500

Protein production:

S. No.	Protein Expression 5L scale	Price for one batch (USD) (Expression, purification, and QC)	Price for three batches (USD) (Expression, purification, and QC)	Price for five batches (USD) (Expression, purification, and QC)
1	Microbial	4,500	12,150	18,100
2	Mammalian	8,000	21,600	32,250
3	Insect cells	6,500	17,550	26,220

- Cell free protein expression is currently not available.

mRNA synthesis (IVT):

S. No.	mRNA generation (IVT)	Price in USD for µg scale (up to 100 µg) (template synthesis, mRNA and purification and QC)	Price in USD for mg scale (up to 100 mg) (template synthesis, mRNA and purification and QC)
1		2,500	22,000

*mRNA size assumed as 600 base pair.

AAV generation:

S. No	AAV generation and QC	Price in USD 1e12 VG (ultra centrifuge purified)	Price in USD 1e14 VG (ultra centrifuge purified)
1		5,000	20,000

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11. Annexure 3

Rate card for Oligo FFS services:

For regular 3-10-3 gapmer : Any other Oligo sequence or significant changes in the monomer will have different price and will be implemented on agreed price.

Deliverable	Price per Oligo (USD)
1 - 2 mg, <300 compounds	310
1 - 2 mg, >300 compounds	300
5 mg, <50 compounds	580
5 mg, >50 compounds	520
25 - 30 mg, <10 compounds	1,800
25 - 30 mg, >compounds	1,750

Raptamer:

Compound type	Synthesis Scale	Purification technique and purity	Price per compound (USD)
Raptamers (56 mer) without any modifications at the 5' or 3' ends	1 µmol	Desalting : >80% purity	$175
Raptamers (56 mer) without any modifications at the 5' or 3' ends	1 µmol	HPLC; >90% purity	$240
Raptamers (56 mer) with 5'-biotin modification	1 µmol	Desalting: >80% purity	$240
Raptamers (56 mer) with 5'-biotin modification	1 µmol	HPLC: >90% purity	$310
Raptamers (56 mer) with a 5'-fluorophore modification	1 µmol	HPLC; >90% purity	$310
Raptamers (56 mer) with a 5'-fluorophore and 3'-biotin modification	1 µmol	HPLC; >90%	Will be discussed later
Raptamers (56 mer) with a 3'-inverted dT modification	1 µmol	HPLC;	Will be discussed later

For regular 18 – 22 mer:

Compound	Supply scale	Purity	Price (USD)
ASO	15 – 20mg	HPLC: >90% purity	2,050 – 2,150

Any other Oligo sequence or significant changes to the Oligo will have a different price. In such case, APSL will assess the request and share a separate quote.

12. Annexure 4

Additional hourly rate applicable for specific instrument usage:

Instrument	Hourly rate in USD
SPR (Biacore) & Octate	100
High resolution mass spec	125